Aral Cimcim

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Portfolio & Projects | GitHub

Profile

Machine Learning Engineer focused on molecular property prediction, generative AI, and computational pathology. Extensive experience in geometric deep learning, deep reinforcement learning, and agentic engineering. Experience in integrating vision foundation models and developing usable AI applications in the medical sector (RAG, deployment, and software engineering).

Selected Projects & Results

Molecular Toxicity Prediction – Competition Winner 2026

JKU – AI in Life Sciences

- Attention-based Graph Neural Network for 12 biological assays (∼12,000 molecules).
- Achieved ROC-AUC: **0.82** – Competition win on Tox21.

Chemprop-Conditioned Diffusion for Multitask Bioassay Prediction 2026

Bachelor's Thesis, Supervision: Dr. Sohvi Luukkonen (JKU), Dr. Johannes Schimunek (University of Cambridge)

- Investigated conditioned diffusion models vs. Graph Neural Network baselines for molecular property prediction.

Vision-Language Models for Pathology Reports 2026–Present

Research project, Supervision: Prof. Erich Kobler (IML JKU), Dr. Sabina Köfler (Institute for Molecular Pathology)

- Development of an automated tool for generating structured pathology reports from whole-slide images using foundation models with explicit reasoning.

TrialTrace & Molescope 2025–2026

Software engineering

- Interactive RAG application for clinical trial protocols and molecular evidence explorer (ChEMBL, Tox21, ESOL) with chemical and semantic search.

Practical Experience

AI / Computer Vision Internship 2025–2026

JKU – Computer Graphics, Computational Imaging, Machine Learning, Intelligent Optics, Prof. Oliver Bimber

- Development of a custom user interface and processing pipeline for optical cross-section imaging workflows.
- Integration of Vision Transformer models (VGGT, DA3) into a unified end-to-end field-deployable processing pipeline for Austrian firefighters.

Education

BSc Artificial Intelligence JKU Linz – Ongoing (2023 - Expected 02/2027)

Relevant Coursework: LSTM & Transformers, Computer Vision, Geometric Deep Learning, AI in Life Sciences.

MA TransArts University of Applied Arts Vienna (2022)

Technical Skills

Programming	Python, R, MATLAB, C, Mathematica
Generative AI & LLMs	Vision-Language Models, Transformers, Diffusion Models, Hugging Face, RAG
Machine Learning	PyTorch, scikit-learn, TensorFlow/Keras, PyTorch Geometric, RDKit
Deployment & Tools	Docker, Microsoft Azure, Git, Weights & Biases, Ray
Data & Evaluation	SQL, statistical analysis, classification metrics, uncertainty estimation

Certifications & Languages

Oxford Machine Learning Summer School (OxML) 2024 – Representation Learning and Generative AI

German C1 | **English** C1 | **Turkish** Native